MET Boston University

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Study case:
Male vs Female

Is there a wage gap?
What are the determining factors of salary?

AD 688

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Project Overview

Gender Disparities in Hiring and Politcal Influence

Objective

Analyze gender disparities in employment, representation and wages across industries, occupations and U.S. states

Key Questions

- How much of the gender wage gap can be explained by occupation and industry?
- Does gender remain important after accounting for other factors?
- Where are women underrepresented in the workforce?
- Does political geography have influence on the wage disparity?

Data Overview

1.57 M
Employed Workers

72k +
Job Postings

3.2 M
Person Records

Primary Dataset

Integrated Public Use Microdata Series (IPUMS USA)

Coverage: 2024 American Community Survey

Use: Workforce representation, wage gap analysis, clustering and machine learning models

Why it matters: Provides detailed worker-level data needed to measure gender disparities in employment and earnings

Secondary Dataset

Lightcast Job Postings Database

Use: To validate workforce trends against current labor market demand

Why it matters: Connects workforce outcomes to real-world hiring demand

Pipeline Architecture

1. Extract

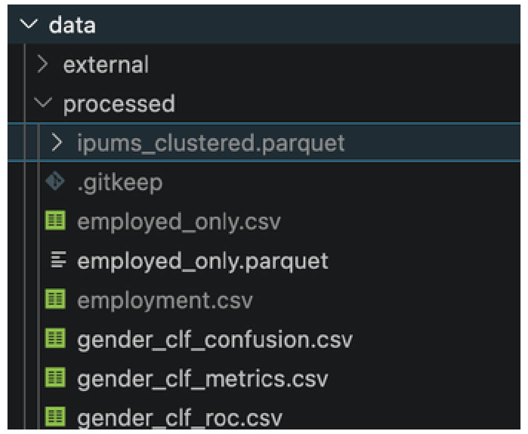
IPUMS: 2024 ACS, 3.2M records
Lightcast: 2024 job postings 72.000

2. Transform & Clean

IPUMS: 1.57M employed, positive wages
Lightcast: 72.000 postings cleaned (salary, state, NAICS)

3. Load & Features

/data/raw
to
/data/processed

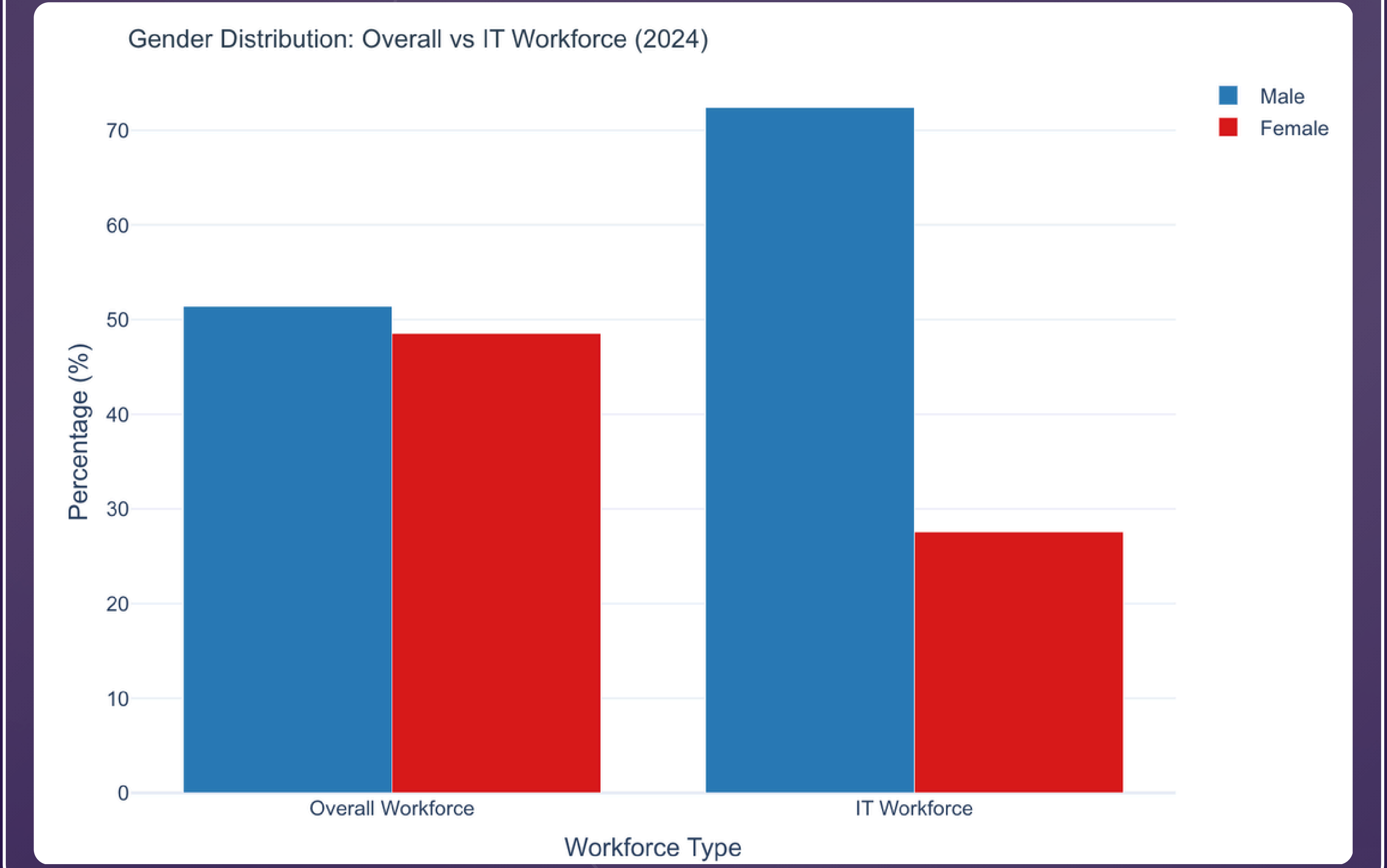


4. Model & Serve

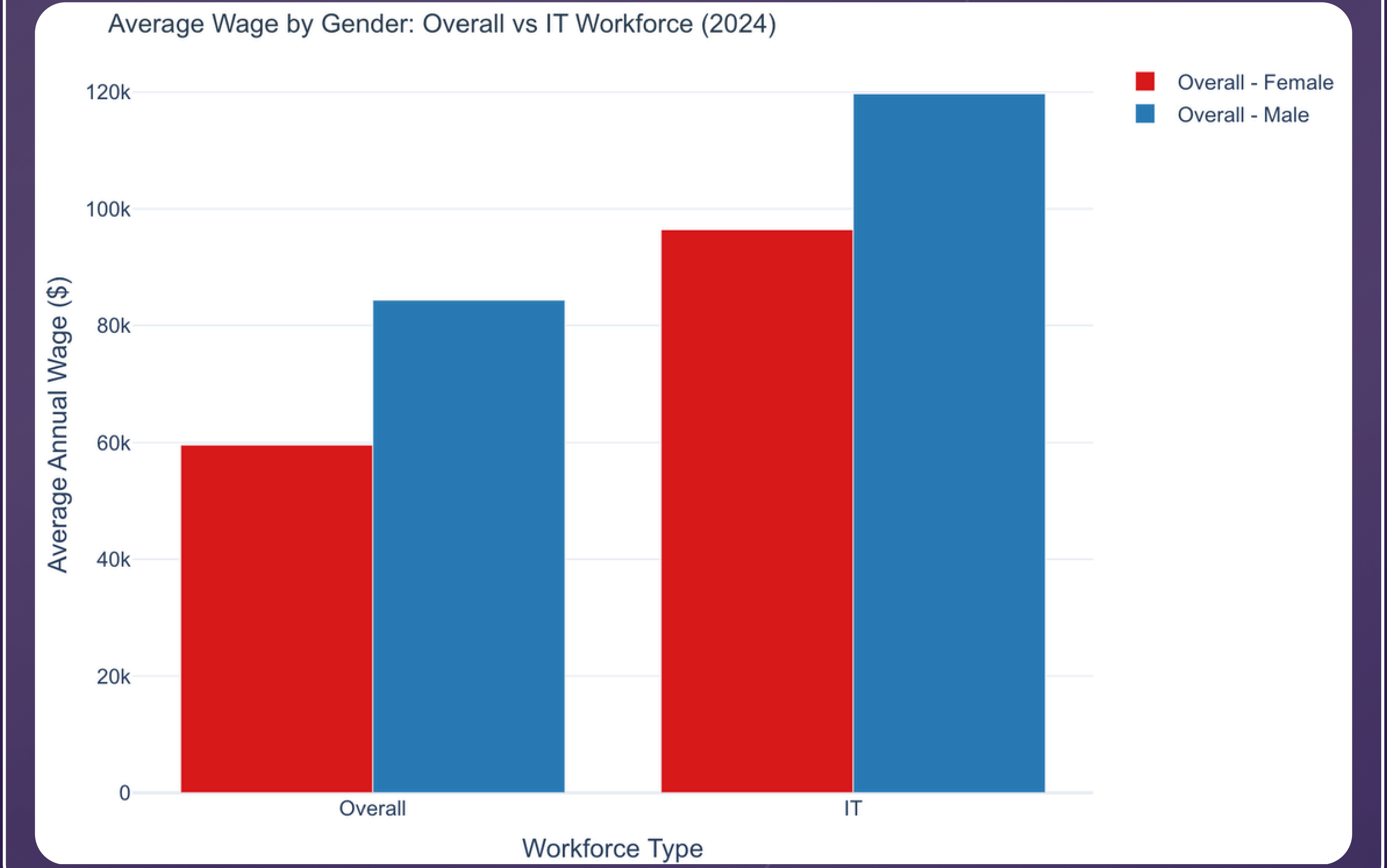
Cluster + Classification
on the IPUMS side,
→ joining the Lightcast job posting to see wage by gender across states.

Visuals

Gender Distribution by Workforce

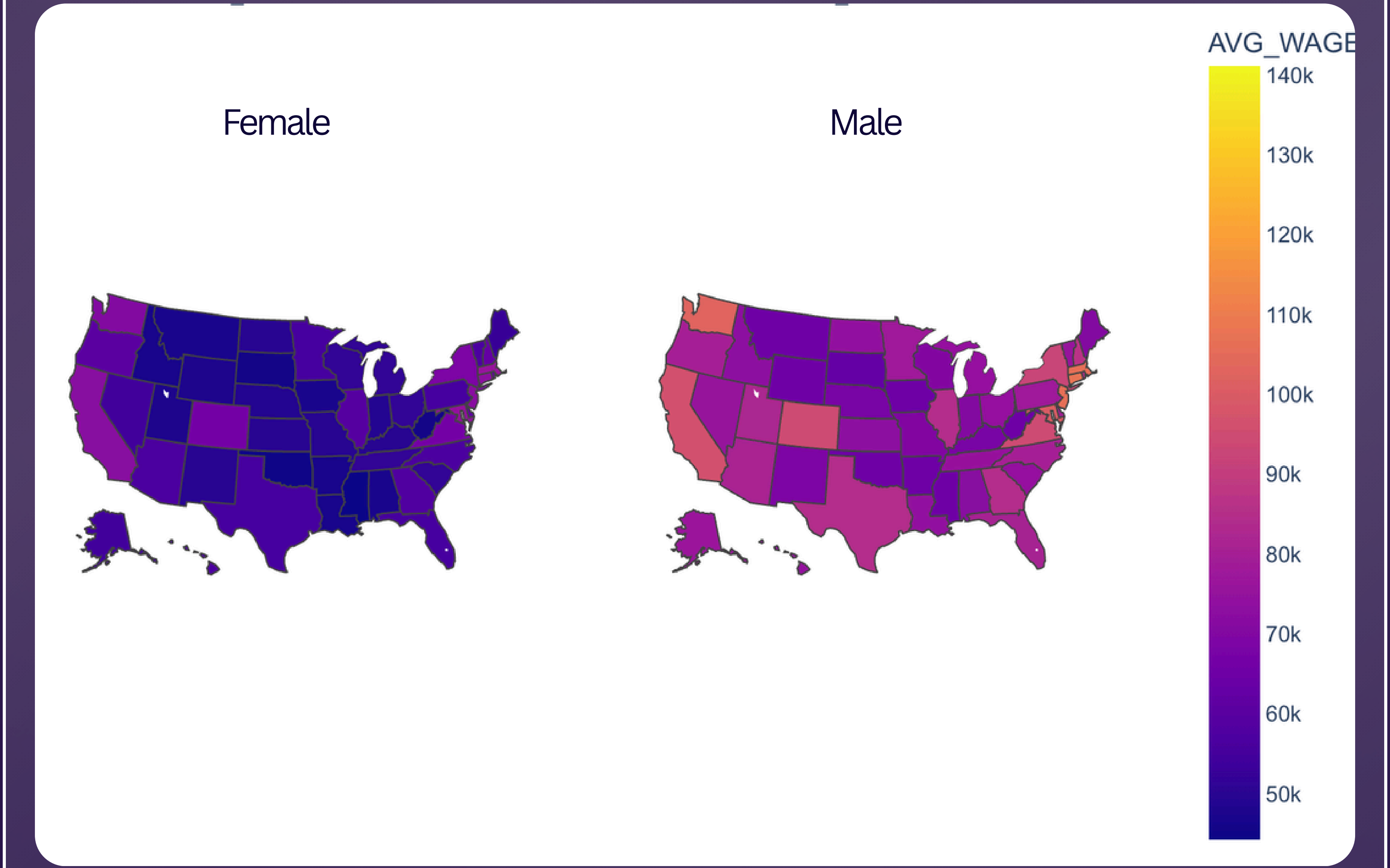


Average Wage by Gender

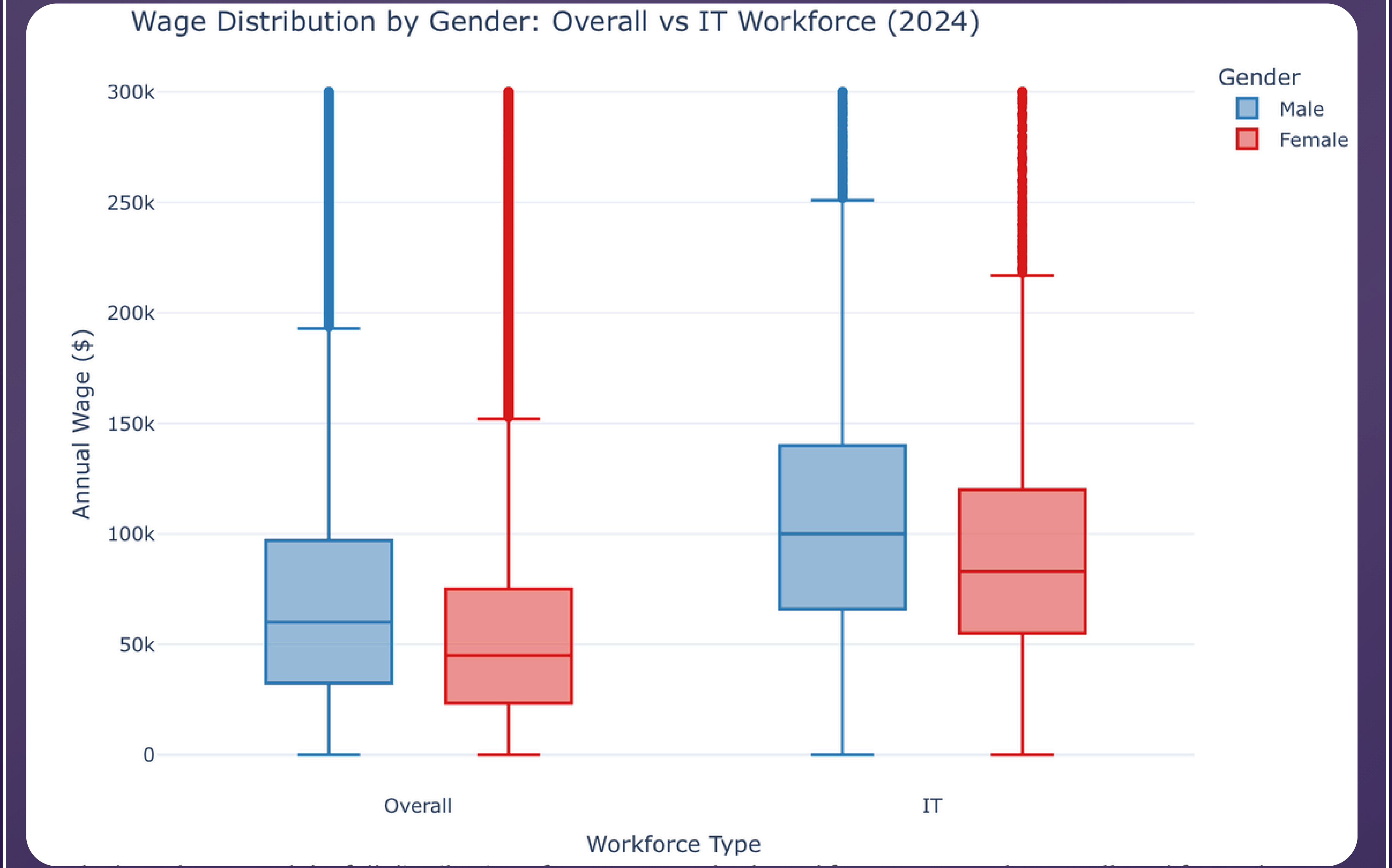


Visuals

Average Wage by State and Gender

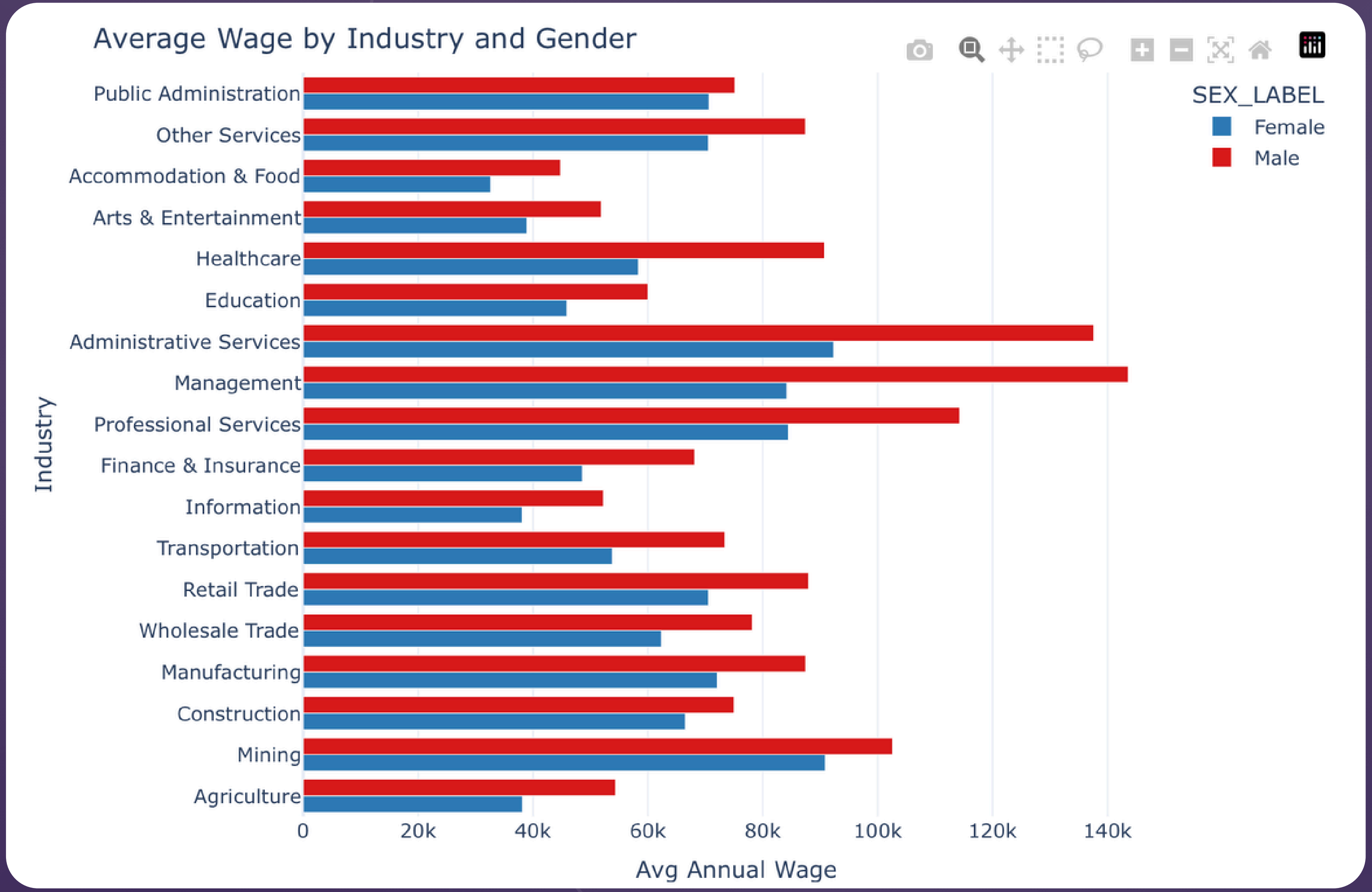


Wage Distribution by Gender

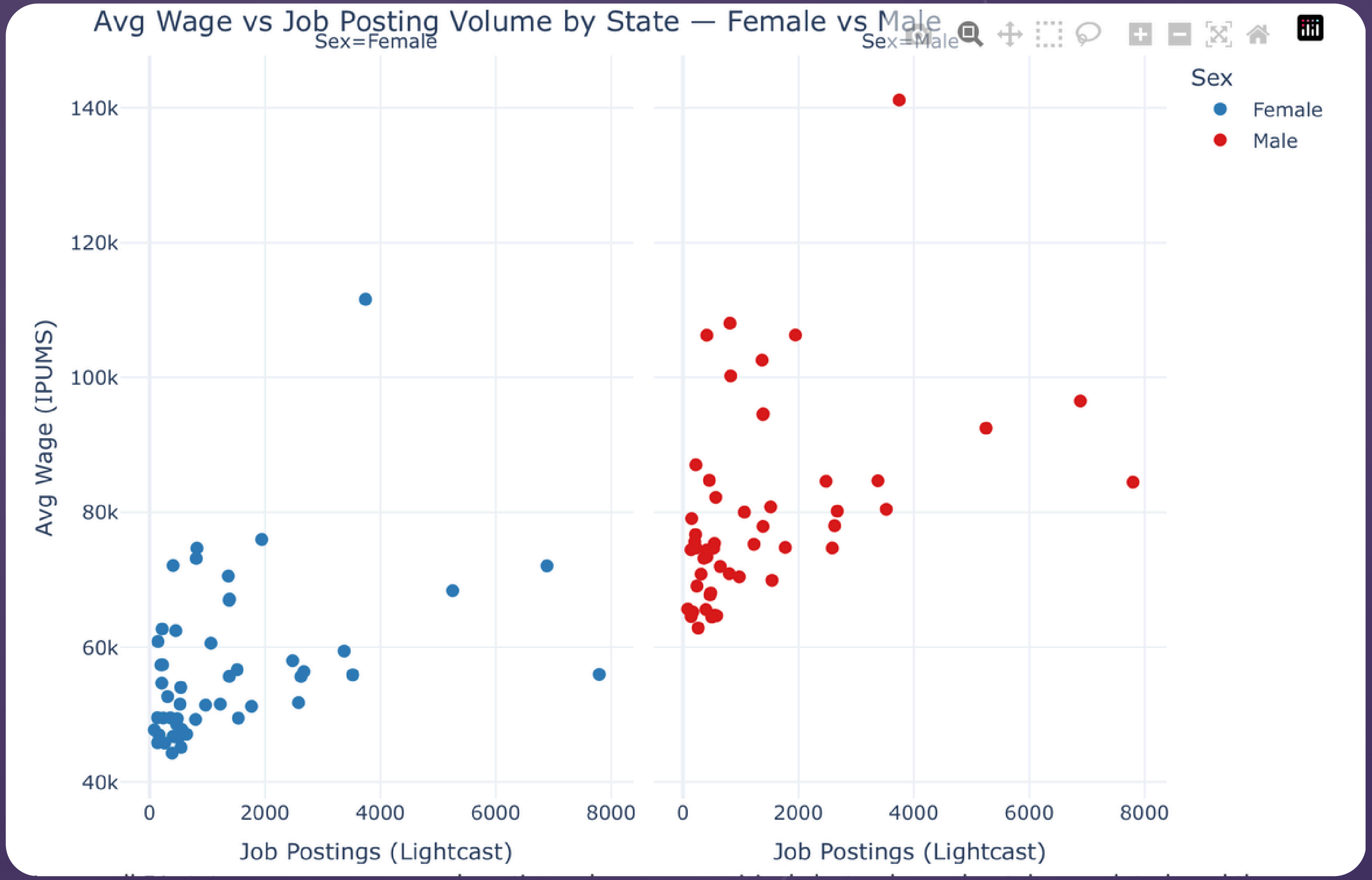


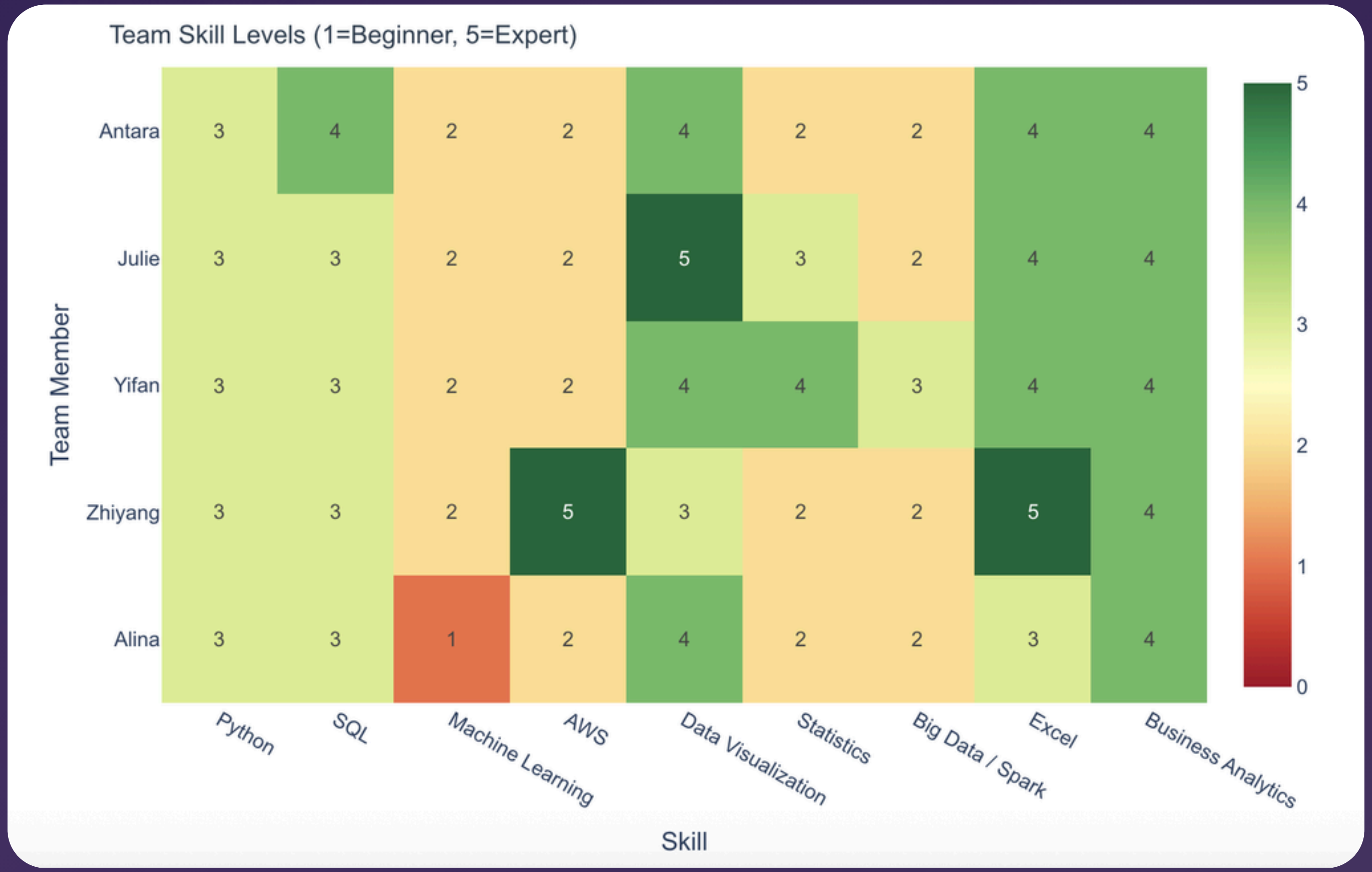
Visuals

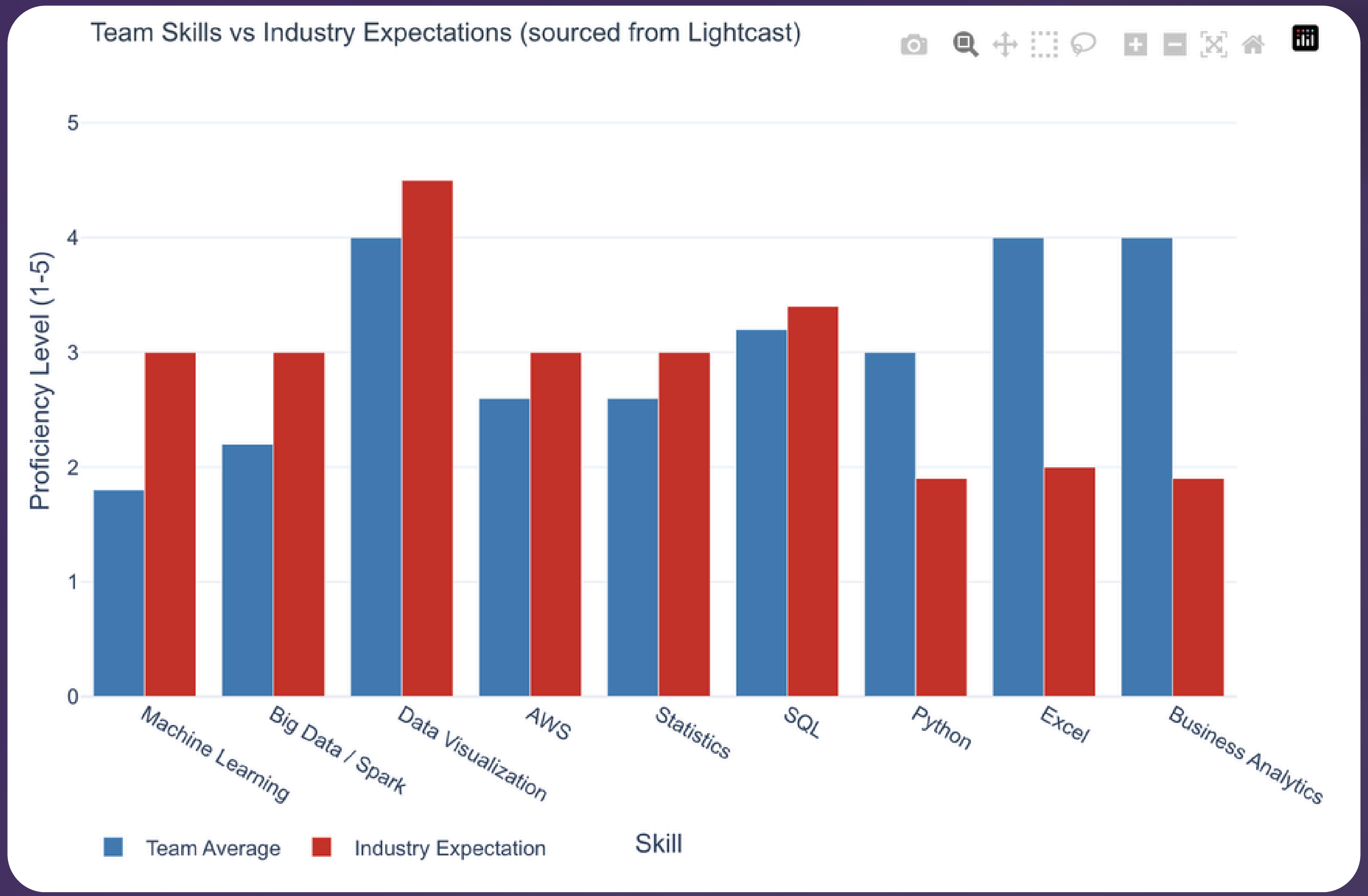
Average Wage by Industry and Gender



AverageWage vs Job Posting Volume by State







Skill Gap Analysis

Comparing our team's current capabilities with 2024 Lightcast IT market demand

Key Findings

- Team average meets basic requirements in SQL, Excel, and Business Analytics
- Strong specialization in AWS and Visualization
- Largest gaps exist in:
    a. Machine Learning
    b. Statistics
    c. Spark / Big Data
- Soft skills like Communication and Problem Solving remain highly demanded in the market

Machine Learning

Clustering + Classification on IPUMS

KMeans (unsupervised) groups workers by wage and age into four pay tiers

Logistic regression predicts a worker's sex from age, wage, occupation and industry.

the female share falls as pay rises, from about **49%** → **23%** in the elite tier

23%

Logistic ROC AUC: **0.83**

gender is recoverable from a worker's economic profile alone.

Linear Regression Decomposition

Raw Gap

$24,749/year

Controlling for age, race and state only

Adjusted Gap

$22,426/year

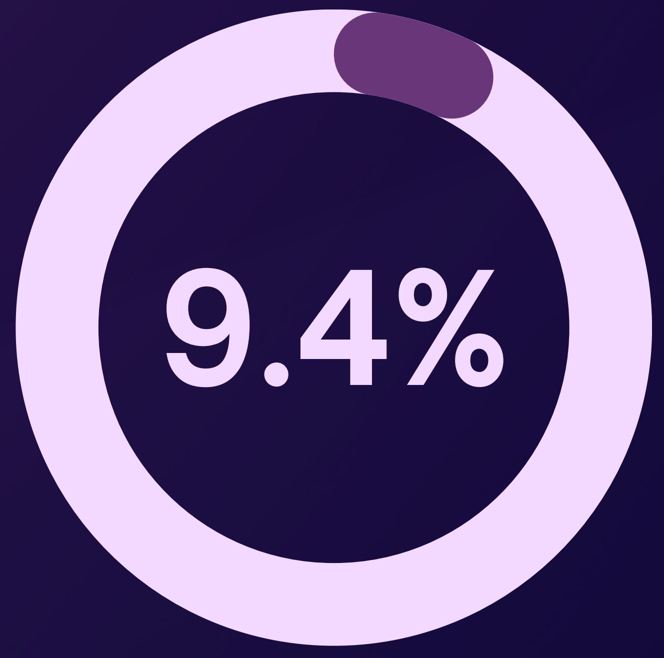
Adding occupation and industry

To quantify exactly how much of the wage gap is explained by occupation/industry vs. unexplained

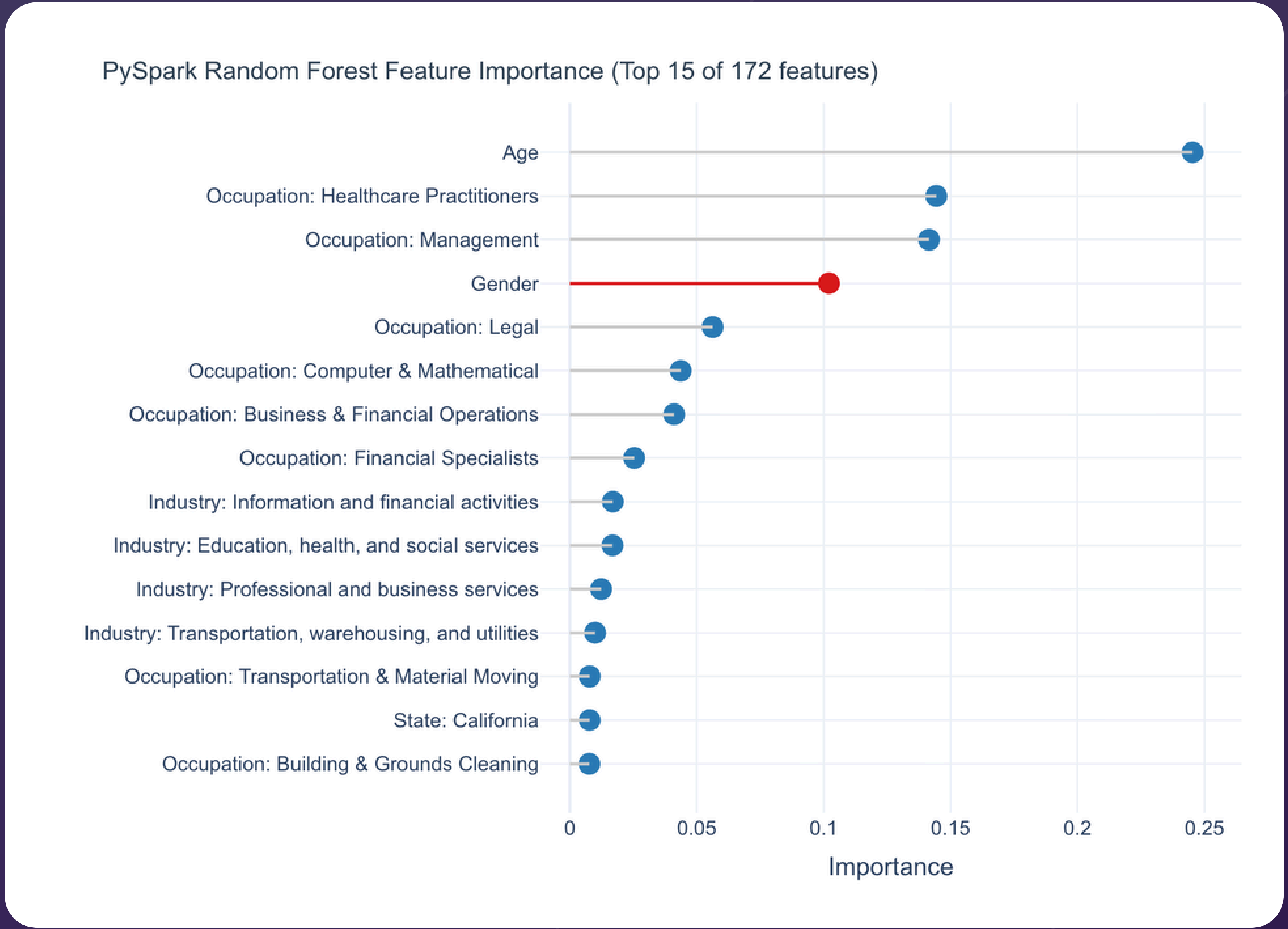
Baseline model: Age + Race + State + Sex

Conclusion: The gender wage gap is not primarily a story of women choosing lower-paying jobs

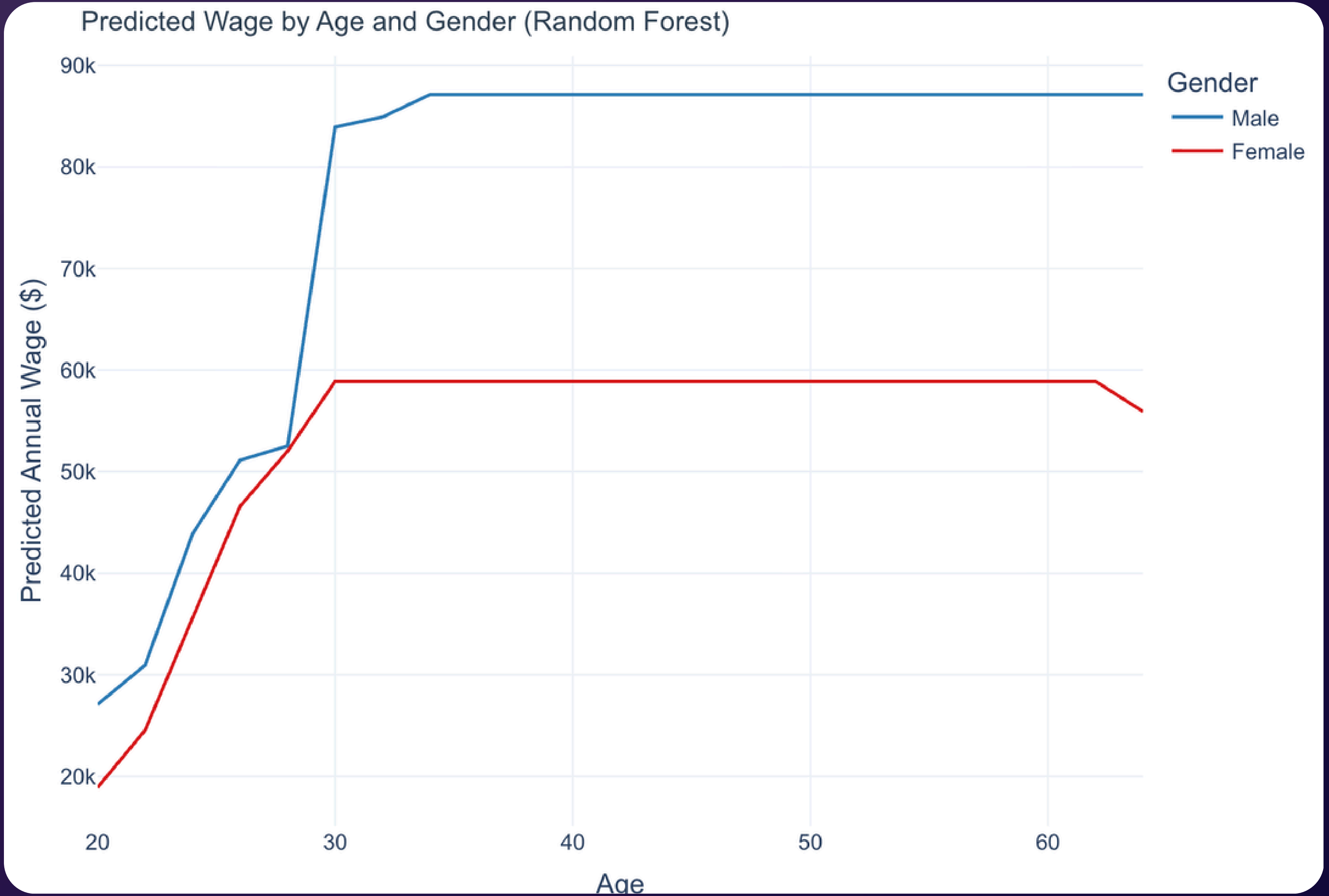
Unexplained



Explained



Random Forest



100 trees | Max depth 10 | 200K row sample

Parallelism : 2 cores | Partitions : 8

Train R^2 : 0.270 | Test R^2 : 0.236

Key Finding

Even when the model has access to occupation, industry, state and race - gender still ranks #4 of 172 features. This is not a regression artifact; it is a pattern the data independently discovered

Conclusion

83%

0.83 ROC AUC: sex predictable from economics alone. National pattern: 25% (blue) vs 22% (red).

91%

of the gender wage gap is NOT explained by occupation or industry .

raw gap ~$24,700/yr

23%

women in elite-pay tier

VS

77%

men in elite-pay tier

Blau & Kahn (2017)

Sources: Blau & Kahn (2017); Goldin (2014); Arulampalam, Booth & Bryan (2007); Kleven, Landais & Søgaard (2019).

Thank You

Questions?